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Designing Rolling-Horizon Planning for Forest Operations Under Uncertainty

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ABSTRACT
Forest operations operate under tight operational constraints and substantial uncertainty from weather variability, machine availability, disturbances, and market conditions. Rolling-horizon optimization is widely used in this context to balance near-term decision commitment with longer-term adaptability, yet planners have limited guidance on how far ahead to plan or how frequently to re-optimize schedules. In particular, the effects of planning horizon length and re-optimization frequency on schedule quality and stability under uncertainty are poorly understood. This study evaluates the performance of rolling-horizon machine scheduling in forest operational planning using the Forest Harvesting Operations Planning System (FHOPS), an open-source machine scheduling tool for forest operations. Using controlled, semi-synthetic datasets representative of British Columbia forest operations, we systematically examine how rolling-horizon design choices influence scheduling outcomes across varying levels of operational uncertainty. Results show that no single planning horizon dominates across all conditions. There are performance trade-offs between longer planning horizons, which improve coordination across machines and blocks but increase exposure to forecast degradation, and shorter horizons, which are more responsive to change but may reduce overall quality and feasibility. We identify ranges of planning horizons that produce stable schedules and show when different horizons converge to similar outcomes. Frequent re-optimization does not consistently outperform longer planning windows, with performance depending on factors like disturbance frequency and reliability of short-term forecasts. These results provide evidence-based guidance for configuring rolling-horizon planning in forest machine scheduling and establish practical performance bounds under controlled experimental conditions. The open-source FHOPS framework supports reproducibility and adaptation to operational environments, offering a foundation for future applied testing and implementation.

KEYWORDS
forest operational planning; rolling-horizon optimization; simulation; machine scheduling; decision support

Introduction

Forest operations have become increasingly complex with advances in harvesting technology and expanding expectations for sustainable forest management. As a result, decisions concerning the timing and allocation of harvesting equipment must satisfy tight operational constraints while trying to stay consistent with tactical and strategic

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planning objectives (Martell et al. 1998). Delays in deploying equipment can disrupt the supply chain and reduce available timber volume for mills (D'amours et al. 2008; Alayet et al. 2013) as well as increase operational costs, particularly in operational environments characterized by uncertainty from weather variability, machine breakdowns, disturbances and market fluctuations (Arora et al. 2023).

Sustainable forest operations are the critical day-to-day implementation of broader forest management plans (Marchi et al. 2018), and poor planning at this level can have cascading effects across planning levels, undermining long-term forest management objectives (Myers 2004). As operational environments continue to evolve with climate change, increasing intensity and magnitudes of disturbances and expanding expectations for ecosystem service provision, forest operations needs to adapt to greater uncertainty and complexity (Tampekis et al. 2024).

Machine scheduling is a central decision problem in forest operational planning and involves the allocation and assignment of machines and workers to specific tasks in harvest blocks over discrete time steps (Corner and Foulds 2005). In practice, planning approaches vary widely. Small- to medium- sized operators often rely on manual methods such as spreadsheets and whiteboards and experiential knowledge. Larger operations tend to use proprietary commercial optimization software or consultant-developed tools tailored to their specific operational environments. While many of these solutions perform well locally, they are often inaccessible or unreproducible, limiting transparency, broader evaluation and adaptation to new contexts, contributing to fragmented progress in forest operational planning research (Jaffray et al. 2025).

From an operations research perspective, forest scheduling problems are large-scale combinatorial scheduling problems (Bellavenutte et al. 2020). Exact optimization over long temporal and spatial horizons is often computationally intractable. The stochastic nature of operational environments further reduces the practical value of long-term deterministic schedules (Sethi and Sorger 1991). To address both computational and informational limitations, rolling-horizon optimization is commonly used to decompose full-horizon problems into solvable sub-problems that are repeatedly re-optimized as new information becomes available (Venkataraman and Nathan 1999).

Rolling-horizon planning is particularly beneficial in forest operations, where near-term decisions need to be firm but longer-term plans act primarily as guidance (need not be mathematically stable) to support contractor risk management while adapting to evolving conditions (Feng et al. 2010). However, the effectiveness of rolling-horizon approaches depends on design choices like planning horizon length and re-optimization frequency.

In hierarchical forest management planning, limited anticipation of tactical constraints has been shown to improve coordination between planning levels, although these benefits diminish as anticipation horizons extend beyond the time frame in which operational conditions can be forecast with reasonable confidence (Beaudoin et al. 2008). Forecasting errors tend to accumulate over longer horizons, increasing schedule instability and reducing reliability (Chand et al. 2002). There exists a trade-off between improved coordination through longer planning horizons and increased uncertainty and instability from forecast degradation (Beaudoin et al. 2008). Operations research (OR) theory suggests than an optimal planning horizon exists where marginal forecasting benefits equal marginal forecast degradation costs (Baker and Peterson 1979; Santos et al. 2021; Sethi and Sorger 1991).

Rolling-horizon approaches are widely used in forest operations, but there is limited guidance on how operational planning length and re-optimization frequency affects schedule quality and stability. The tendency of managers to select arbitrary planning horizons can reinforce multiple challenges within the planning process, including high computational burden and limited reliability for longer-term forecasting (Chand et al. 2002). The performance implications of these design choices are unclear.

To address this gap, this study uses the Forest Harvesting Operations Planning System (FHOPS), an open-source forest operational machine scheduling tool (citation), as an experimental platform to systematically evaluate rolling-horizon planning performance. Using controlled, semi-synthetic datasets representative of British Columbia forest operations, we examine how rolling-horizon design choices influence schedule outcomes. The objective of this study is to provide evidence-based guidance for practitioners on planning window selection and re-optimization strategies for operational forest machine scheduling. While specific recommendations will depend on local operational characteristics, the underlying patterns would reflect general properties of rolling-horizon optimization. We address four specific research questions:

(1) How far ahead do planners need to plan before schedule quality degrades?
(2) What rolling-horizon length produces stable schedules?
(3) Do different horizon lengths converge on similar results?
(4) Does frequent re-optimization outperform longer planning windows with less frequent updates?

Findings will establish performance bounds under controlled experimental conditions with real-world testing and validation as an important future step.

Materials and Methods

Study Design and FHOPS Framework

We employed a factorial experimental design to evaluate rolling-horizon performance across systematically varied planning configurations using the Forest Harvesting Operations Planning System. The Forest Harvesting Operations Planning System (FHOPS) is an open-source machine scheduling tool for forest operational planning. The system is a Python package with a command-line interface (CLI) that uses Pyomo, HiGHS for linear optimization solving and multiple meta-heuristics for solving including simulated annealing, tabu search and iterated local search. The objective function minimizes X and Y, including Z. The core functions of the model include build, solve and evaluate.

FHOPS implements a rolling-horizon optimization approach by solving a planning problem for a specific horizon length (H), implementing decisions for the first W weeks (commitment period) and then advancing time by W weeks and re-solving until the complete H has been solved for.

All experiments used FHOPS (version) and SOLVER (ideally HiGHS but also Gurobi maybe if needed) with the following solver parameters: time limit, tolerance, min number of iterations; the experiments were run using [computational specifications].

Data Inputs and Pre-Processing

Base spatial and stand attributes were derived from the provincial Forest Harvest Authorizations layer and Vegetation Resources Inventory (VRI) datasets across five Timber Supply Areas (TSAs) in British Columbia. When overlaid, these datasets provided harvest block geometries, areas, species composition, harvest constraints and volume and density estimates required for harvest system selection and machine productivity estimation. To support controlled experimentation, synthetic variability was introduced into selected operational factors, including operational environment and disturbance level. This experimental approach allows us to isolate planning effects that cannot be tested reliably using historical data alone. The results provide guidance for operational machine scheduling while forming the basis for real-world validation and implementation. Slope classes are derived from the DEM and expressed as an average percent gradient per block. Although the model inputs are deterministic, DEM-derived slope produces a distribution of harvest system applicability within each TSA, validated by the information in the TSA data packages.

Case Study Approach and Scenario Descriptions

A case study approach based on controlled, synthetic datasets was used to evaluate the performance of FHOPS across problem sizes that may be encountered in forest machine scheduling. Three representative datasets were developed to span the range of problem sizes that may be encountered in machine scheduling in forest operational planning. These datasets share a consistent structural form for comparability but vary in the number of harvest blocks, machines and decision variables. The parameters used reflect operationally realistic conditions typical of BC forest harvesting operations, including machine productivity ranges and block sizes. They are categorized as small, medium and large, reflecting increasing computational complexity. The small and medium-sized datasets are solvable using mixed-integer programming (MIP) within practical computational limits and are used to evaluate solution quality, stability and convergence behaviour under rolling-horizon re-optimization. The large dataset tests the scalability of the approach, while the largest dataset exceeds tractable MIP limits and is used to examine heuristic solution response. This is only made for one of the datasets given that we are testing the computational bounds rather than the complexity of the operation itself. With the number of blocks being the dominant factor computationally, should this shit the bed, it wouldn’t matter where it was happening.

Table 1.: Summary of Dataset Characteristics

Dataset	*Blocks*	*Machines*	*Landings*	*Terrain Types*	*Prescriptions*
Small	6	6	2 per block	1	2
Medium	18	12	2 per block	2	3
Large	40	24	2 per block	3	5

Experimental Design

We designed an experimental framework to evaluate the performance and convergence characteristics of FHOPS, using a rolling-horizon approach. The primary objectives are to quantify the bounds of different planning horizon lengths, assess the value of frequent re-optimization, compare convergence between rolling-horizon and full-horizon solutions and evaluate robustness in typical British Columbia operational environments.

All experiments were conducted on both identical and multiple datasets to ensure comparability as well as test out applicability in a range of operating conditions. Full-horizon plans were solved with the MIP (or meta-heuristic, if computationally too big for the MIP) and then these serve as the baseline against which to compare the plans using rolling-horizon decomposition. For each plan, the factorial design was used with the factors presented below.

For each of the sizes of datasets (denoted as k presented in Table 1), we evaluated a series of experimental runs. We evaluated a range of operationally plausible horizon lengths, denoted as θ:

$$\theta \in \{2, 4, 8, 16\} \text{ weeks}$$

Short horizons were included to explore the performance limits of myopic planning, while longer horizons were included to explore the point at which diminishing returns or solver difficulty affect the overall solution. For a set period of time (planning horizon, θ), we will solve the model with a set commitment period or rolling-horizon (f_{roll}) which identifies the frequency of re-optimization as time advances. We considered three frequencies, representing daily, weekly and bi-weekly re-planning cycles:

$$f_{\text{roll}} \in \{1, 7, 14\} \text{ days}$$

Model behaviour can vary across operational environments so to test this out, we assessed multiple operational environments representative of BC:

$$c \in \{\text{Coast}, \ \text{Interior}, \ \text{Boreal}\}$$

These contexts have multiple different factors that can affect machine scheduling and productivity. Including a range of contexts allows us to examine if results are context-dependent and how well FHOPS acts as a generalizable tool across operational environments. Ground-based systems in lower slope blocks and cable- on higher slope blocks (within TSA different conditions based on slope).

The total number of experimental runs is given by:

$$N_{\text{runs}} = K \times \Theta \times F \times C$$

to equal 108 runs.

Key Performance Indicators and Analysis

Performance across the experimental runs was measured using multiple key performance indicators (KPIs). Some KPIs are included in the FHOPS framework; other KPIs were calculated for research. We measured performance across:

- Schedule quality
- Stability (check if *block_id* changed from roll k to k+1 and then compute fraction of unchanged) with rescheduling rate being 1 - stability.
- Computational performance (CPU, solution time, memory usage, success rate (crash or no crash)... reporting on a per problem solved (by size class) basis to give people an order of magnitude with associated reference in discussion (honest of things being slow but not need to apologize)

- Convergence metrics, Optimality Gap (%) between solver objective values of rolling-horizon plan and long-horizon MIP (deviation from benchmark of entire planning horizon with perfect foresight... might be difficult with the big MIPS). For the problems small enough to solve with MIP, how close did we get? Quant and qual descriptions of convergence.
- What is the longest MIP we can actually solve and how does that compare to what we'd like to solve?

For each experimental configuration, FHOPS outputs were downloaded. Available metrics were used, others were calculated. All analyses carried out in Python (version) with code and datasets made available.

- Reporting on ranging of outputs
- The statistical analysis appropriate from a large number of deterministic model simulations that explore a range of different inputs (thorough sensitivity analysis)

Reproducibility and Availability

All datasets, scripts, etc were version-controlled. All datasets are publicly available and downloadable. The software stack used is open-source.

Results

We evaluated FHOPS' rolling-horizon performance across X number of experimental runs that represented typical forest harvesting conditions in multiple British Columbia (BC) operational environments (i.e. coast, boreal and interior). In this evaluation, we assessed the effects of:

- Planning Horizon Length
- Re-Optimization Frequency
- Problem Size
- Context

These are reported using the following metrics:

- How far ahead do plans need to be planned to be good?
- What are the trade-offs of only planning X weeks forward?
- cost vs planning horizon length to show how quality improves but stabilizes after a certain point
- Mathematically and computationally what are the limits?
- Figure: x horizon length and y avg cpu time or memory to show limits of large-horizon optimization
- How does solver performance change between planning horizons?
- Is the optimality gap with larger plans larger than we should have? Does it make for poorer plans?
- Diminishing returns to show practical planning horizon lengths (horizon vs performance pareto curves)
- Figure: Pareto Curve (stability, computational cost) to show myopic and hyperopic challenges
- Stability (check if *block_id* changed from roll k to k+1 and then compute fraction of unchanged) and plot against f_{roll}

- a 3 axis graph could be cool
- Looking at horizon length, then re-optimization frequency and then both of them together
- What is minimal bounds of looking ahead to be able to report?
- **Rather than separating out problem contexts and size, we could present them on the same plots as above**
- Table: Problem characteristics across BC contexts (different TSAs)
- Figure: Box plots showing differences across BC contexts (different TSAs)
- Figure: Context-specific effective horizon lengths, what is the best horizon length depending on the factor? Compounding of compute time if horizon length gets bigger then oh no. Measuring quality of plans as function of horizon length (bad to speed to plateau to heuristic could stb) –> Y: stability (goodness) and X: planning horizon
- Solver difficulty vs problem size (# of variables, CPU time) of different TSAs

Discussion

Discussion will highlight areas of opportunity to use these mapped trade-offs to communicate forest management decisions out to the public as well as add to the discussion of what 'good' operational plan options look like from an OR lens. This discussion of 'good' operational planning will compare the different solutions from various rolling horizons to answer (if only partly) the following questions:

- Is this model any good?
- What types of operations does FHOPS solve really well (how well?)
- Can this help improve current practice in the real-world? Comparison with how much time manual planning would take to achieve the same results with coordination and iterative readjustment.
- How far ahead do I need to plan for near-optimal schedules?
- If I lock in two-week chunks as compared to one-week or three-week, how much optimality does it change by? How much optimality do I lose?
- What happens if I only plan 4 weeks and re-optimize weekly?

Role of input data as a limitation but also design feature for systematic evaluation

Model Performance and Practicality

The FHOPS model demonstrates that mixed-integer programming optimization can be used to generate operationally feasible machine scheduling plans across a range of representative scenarios in BC. These scenarios varied in operational environment, harvesting system, problem size...

While FHOPS is capable of producing mathematically optimal or near-optimal schedules subject to operational constraints, it is not intended to replace planner judgment. Forest operational planning necessarily involves many factors that lie outside the scope of formal optimization models, including social relationships, local knowledge, stakeholder considerations, rights-holder consultations, and changing site conditions. The value proposition of FHOPS is in its ability to handle the computational complexity of scheduling decisions, freeing planners to focus on higher-level ecological, societal and strategic considerations that are essential to sustainable forest management.

FHOPS also allows rapid exploration of solution space to create a rational basis to qualify how good an operational plan is. There is now a tool that provides a basis for saying that something is good because now you have something to compare it to. FHOPS doesn't have a bad day (consistency in the computational approach and planning)... ability to ratchet quality up over time. This capability lowers the barrier to testing new planning approaches and reduces the perceived risk (de Pellegrin Llorente et al. 2023) with changing established practices.

Across the tested scenarios, FHOPS was able to solve problems of varying size within computational time frames that are reasonable from a practitioner perspective.

- Acceptance of some level of suboptimality in exchange for being able to plan further ahead without locked in commitment but reasonable forecasting effort
- Different users on the landbase would likely need to solve different planning questions especially considering economies-of-scale for machine scheduling
- Practical use for planners and control to still edit some of the plan without taking away feasibility (for example, adding productivity)
- Emphasis on reasonable forecasts
- Solver speed improvement comparison against real operational
- What tech do ops planners have that can tackle reasonable problems? What is in the order of magnitude for what people would usually have? Rolling horizon parameters with medium to small solutions which we think is pretty good
- Enabling accurate scenario modelling and analysis of the counterfactual without incurring implementation costs
- How do different users want to solve different planning questions?
- Initial advancements into stochastic modelling with more work to be done
- Limitations as recognized the SIMANFOR team with adoption of software in forestry (Bravo Oviedo et al. 2012)
- Using SWOT analysis, this would inform the "what needs to happen before this can be helpful to people" component.
- Time to solve compared to what operational planners would like/benefit from and then what others would actually not complain about. This took 8 hours but it worked out is pretty reasonable. More time to focus on more social aspects of the planning process, allowing the computational (less nuanced tasks) to be done by the computer, freeing up more time for more time important tasks like relationship-building
- testing of the counterfactual without incurring implementation costs
- by providing an accessible framework, organizations can track decision-making processes over time, learning what works and what doesn't in different scenarios.
- By enabling foresight, the modelling framework can link short-term actions, such as equipment scheduling and harvesting, with long-term sustainability and profitability targets, bridging traditionally siloed planning levels.
- The machine scheduling model was evaluated by using multiple scenario datasets that represented typical forest harvesting conditions in British Columbia, Canada. These scenarios ranged in terrain, equipment used in harvesting and size of the problem, allowing us to test the extent of the model, the different solution options and operational feasibility. The size of the problem can vary in both planning horizon and number of blocks to schedule.
- These different parameters of our scenario datasets allow us to test the model's ability to solve operationally feasible problems while also testing the mathematical and computational limits of the model (or what would be acceptable for a practitioner). This helps us provide technical parameter guidelines to users for

adoption and implementation of the technology without them just shooting in the dark for the first couple times... we need it to be useful overall.

- Ability to override default productivity and cost parameters for solving.
- Solver availability and cost (can be worked around with different options, i.e. not everyone having to buy GUROBI but if website, then everyone can use it with only one time fixed cost). Where do the MIPS fail? Where do your usual forestry problems (realistic forest operations use cases for the software) fall?
- FHOPS works well for operations with X characteristics but struggles when Y conditions exist (limit recognition)
- Mixing harvest systems as compared to pure. Assigning systems to blocks as mathematically separate sub problems or sub optimizations, or as one problem
- The model has an identified objective and is subject to various constraints. It is operational. Different signals, either top-down or bottom-up, have been used in mathematical modelling of forest resources hierarchical planning as informed by (Schneeweiss 2003). Beaudoin et al. (2008) explains this approach when they employ a modelling framework to tackle a two-fold challenge of both tactical and operational decision-making, utilizing the Schneeweiss (2003) approach to the wood procurement planning problem by combining tactical and operational levels in a distributed decision-making framework.

Have to address the software sustainability, transparency and reproducibility concerns identified in the other paper (Jaffray et al. 2025)

Planning Horizon Length and Rolling-Horizon Trade-offs

If capable of solving all in one go, no point in resolving further if you can already do it in one go. We know the model struggles to solve the long 16 week problem but do we need to if solving a four week problem can do the reasonable enough work (better quality, lower gap)

Comparison to Current Operational Practice

- Using optimization to generate a strong baseline schedule and then workflow soft constraints can adjust based on factors that the model can't capture
- Operationally, there are also business related benefits that support the adoption of rolling-horizon decomposition. Contractors and operations plan extended horizons for risk management and business coordination that don't necessarily require mathematically optimized solutions. Past firm commitments of crew assignments, machine positioning and work guarantees, rough forecasting further out can support contractor planning, maintenance scheduling, budget management, etc even if they don't know the nitty gritty block details or which machine is going to be having a bad day or whatever but the rough plan still helps. The math doesn't need to be locked down that far ahead anyway as we all know that operations can change quite quickly but there needs to be a reasonable plan at least.

In practice, operational planning horizons are determined how? For each planner, thoughts could range from "we plan this way because we've always done this" to "we don't know mill demand 4 weeks out, why are we optimizing 16 weeks ahead?" to "we have no way to test it without manually rebuilding all the schedules and possibly losing operational efficiency to try out a new option, too risky". Answering these thoughts is easier when you have an evaluation framework that can be tested quickly across different operational environments and conditions.

We can't really give you a strong statement on how good or bad this is or how actual operational plans are good or bad because we don't actually have the data.

(Morton and Pentico 1993)

FHOPS provides mechanism of baseline to be adjust by planner judgment. Comparison to manual planning approaches (time-intensive, limited number of scenarios that can be reasonably evaluated). If we can show that FHOPS works well in a hypothetical world, then maybe you can afford to replan much more often than you usually would. Freeing up more time.

- Planning horizon of three months (one season) or 90 days
- Shift-level decision-making (option for daily or hourly decision-making if changes to model/evaluator)
- Variation in planning horizon
- Why reasons for planning 16-weeks out (forecasting, not committing) is mainly out of risk management for the contractor, wanting to know where they would be going but mathematically, what are the limits of where we need to plan? I.e. the bounds of the planning horizon - interest in having a longer scheduled plan is outside of needing it to be mathematically optimal or stable - a shorter model could likely do better than the guy with Excel and it would get that solution much faster, allowing you to more quickly explore the solution space

How far ahead must managers plan to achieve effective outcomes? How short can we plan to be effective? Planning horizons vary substantially in the operational realm of hierarchical forest management. Why are people planning on their current planning horizons? Is it just for know or how did they decide to do this? What are the limitations of planning too myopic or hyperopic? Rolling-horizon optimization, which repeatedly solves smaller short-term planning problems that are advanced through time, is an industry standard that allows planners to respond to the certainty of uncertainty in forest operational planning. However, performance characteristics of these rolling-horizon methods are somewhat limited. Rolling-horizon planning accommodates business realities by providing credible longer-

Rolling-horizon planning can update as information improves which aligns well with contractor business risk management because new info can be incorporated as needed and there is still a fair bit of flexibility which helps us respond to the inevitable uncertainty in forest operations. However, implementing rolling horizon planning requires us to decide how far ahead we should be planning in operations? How far is too far? How short is too short? We would like to avoid both near- and far-sighted decision-making which would give the impression that if there is near and far sightedness, there would be a practical rolling horizon length before one of these occurs that would be beneficial from both a business perspective and also from a math perspective.

The findings of this study support evidence-based selection of planning horizons based on specific operational environments.

Strength of human expertise to react quickly, know the intricacies of the planning problem

(Schweier et al. 2019) (Tampekis et al. 2024) (Ulvdal et al. 2023)

This suggests that an optimal operational planning horizon should be long enough to capture the realities of the operation but short enough to not be reliant on uncertain forecasts. Knowing operations are so variable and site-specific, it is unlikely that a single universal optimal horizon would exist across plans.

Planning too short creates myopic decision-making. Conversely, planning too far ahead creates hyperopic decision-making which can accumulate forecasting errors.

Re-optimization Trade-offs and Optimality

- The optimal re-optimization frequency depends on horizon length. Based on the different planning horizons, how do different re-optimization frequencies affect the quality of the solutions?
- Firm scheduling and actual commitment of resources on a daily (weekly) maybe two week scale for the contractor that is adjusted for weather, breakdown, quotas (even-flow possible?), safety concerns etc
- 1-3 months rough operational planning window that is tentatively planned but can still shift around depending on what is available, what is needed, road access and seasonal constraints, mill delivery needs (gotta feed the client), contractor capacity, block readiness (permitting, engineering, etc) - what is complete and what is held up?
- Reasons why plans are good to have with reasonable knowing includes fibre flow, equipment utilization, fair work distribution, etc so that the contractors can have more certainty from the licensees and the licensees can have more predictability about their timber flow and management on a bit of a longer time frame
- Rescheduling and robustness might be more important than pure optimization for smaller more uncertain operators on the landbase
- Smaller landowners (private etc) or small area-based tenures might need more flexibility so like less blocks and need for optimization being a big problem but the open-source could be helpful to keep costs low especially if they don't have a set forestry staff or someone who can do this as their full job but still want to keep things cost-effective at a smaller scale
- Interaction effects of the factorial experimental design matter. The optimal configuration is not independent but depends on problem size, horizon length and disturbance levels within the operational environment. This further supports the need for site-specific models as acknowledged in Jaffray et al. (2025). There is not a single recommendation of what best practice would be but rather it is more context-dependent and needs to be a feature of the different factors at play within the model... how do I say this more technically and evidence-based? What level of re-optimization is sufficient?

FHOPs models the technical optimization component required for operational planning's core decisions: which blocks to harvest when, what equipment to assign where and how to sequence tasks spatially and temporally. However, real operational planning clearly isn't just a quantitative thing. The model clearly cannot capture that Joe prefers to use one 855E over another or that a block isn't going to be harvested because the farmer needs to have road access that specific week for his daughter's wedding but like how to express these vibes academically? These aren't model limitations but we need optimization (finding best solutions to well-defined problems) to act as the computational baseline before we can even begin to integrating quantitative optimization with qualitative judgment, stakeholder management, rights-holder management that we need for good operational planning. This baseline can then be adjusted by planners based on factors that the model cannot capture.

- Alignment with contractor and budgeting cycles
- More flexible with credible forecasts but specific details not locked down (i.e. not having to legally commit resources, rates, etc) whereas full optimization might imply more certainty than the business can actually provide
- Rolling-horizon planning has more frequent updates to allow for feedback loops, less time to lock down larger schedules, incorporates uncertainty that can exist around stumpage, fuel, markets, mill demands, etc (volatility) and new information (weather, equipment changes, etc)
- Forecasting errors are worse over long periods of time whereas smaller timesteps would be more robust

The findings of this study reinforce the argument that strict mathematical optimality is not always the most effective criterion for forest operational planning. As emphasized by Beaudoin et al. (2008), anticipation and an understanding of the limits of optimality are essential when applying OR models to operational decision-making. In highly uncertain and dynamic environments, robust and feasible solutions may be more valuable than optimal solutions that rely on fragile forecasts (Myers 2004). Based on this logic, FHOPS is a decision-support system that allows planners to understand trade-offs and explore counterfactuals.

This tool supports the integration of quantitative optimization with human expertise that is essential in the forest operational planning space.

- Uncertainty accounted for generally better than optimality
- Steps away from optimality may offer more robust responses
- Plan that can be audit-able and checked out by the users (i.e. not black box solver) better for trust
- Operational feasibility over optimality with comparisons between MIP and heuristic solutions showing better response to robustness ((Myers 2004)) –> did we need to go to heuristics, yes or no? Is this better or worse than a dude with a spreadsheet?
- Robustness over optimality, maintaining performance in the face of different disturbances so its more robust and useful
- Computational feasibility
- How do these compare with the MIP? Here, introduce Beaudoin et al. (2008) and discuss anticipation and the limits of optimality

Study Limitations

- Largely deterministic model with opportunity to improve stochastic additions and impacts to modelling solutions
- FHOPS does include advanced modelling functions that were not included in this scope of analysis, hyperparameter tuning, stochastic playback, disturbance levels inclusion...
- Measure of how disturbance affects robustness of solutions could be option... we could conjecture that heuristic solutions allow measurably suboptimal would be better in robustness (a likely important quality of real forest operations, conjectures for further research)
- Future work should extend this framework to ....
- What is missing from the scope of the model?
- Budgets (realistic computational capacity)

- Semi-synthetic datasets gave us control over the experiment while still being operationally realistic. Easiest best datasets for writing the purpose of the thesis, made with VRI data that was compiled with more strategic planning details in mind (not a reliable operational basis)
- With much higher input data at the stand level, there would be more details that could be used in operational planning
- Disturbance scenarios tested vs real world complexity
- This research evaluated a series of disturbance scenarios that are simplifications of real-world complexity
- Difficulty with forest operations people and more software engineering focused approach, easier to use than non-existing software
- Project sustainability
  - The project is in-house by the FRESH Lab (Github) and remains in their control with invested interest by the PI (user documentation is clearly outlined in the readmydocs)
- Opportunities with modular design to improve
  - Contract basis if needed for specialized more gatekeeping approaches
  - There is value in being able to plausibly simulate forest operational planning.
- Discuss the replication design and how it allowed us to look at planning decisions instead of random chance or variation.

Conclusion

This study evaluates the rolling-horizon performance bounds of the FHOPS tool in forest machine scheduling. Rolling-horizon planning does not need to mimic full-horizon optimization to be effective.

- Brief synopsis of findings with reference to planning horizon length, re-optimization frequency and disturbances (robustness test) in typical BC operational environments
- Computational feasibility is acceptable for small-to-medium operations, large operations may require commercial-grade solvers? Something kinda like this sentence to frame scope of what FHOPS application can be for user type, it can help align scheduling to be math efficient and business credible
- Address RQs

The conducted experiments make it so that we have a good testing ground and background for setting similar parameters in actual operational environments.

These findings support planners' decision-making in deploying FHOPS or similar tools by informing rolling-horizon settings and deployment of FHOPS or similar tools. More broadly, this work demonstrates a reproducible approach for evaluating operational planning tools. Future work should validate findings using real operational data, extend to broader operational environments outside of British Columbia and continue to develop the open-source tools available for forest operational planning.

Contributions

Conceptual model design and framework, data analysis and interpretation, manuscript writing: R.J.. Coding the model: G.P.. Manuscript writing and revision: Kathleen.

Code and Data Availability

All data and code supporting this study's findings are openly available:

- Data and analysis code
- FHOPS software (insert Github URL)
- Semi-synthetic datasets

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An unnumbered section, e.g. `\section*{Funding}`, may be used for grant details, etc. if required and included *in the non-anonymous version* before any Notes or References.

Notes on contributor(s)

An unnumbered section, e.g. `\section*{Notes on contributors}`, may be included *in the non-anonymous version* if required. A photograph may be added if requested.

Notes

An unnumbered 'Notes' section may be included before the References (if using the `endnotes` package, use the command `\theendnotes` where the notes are to appear, instead of creating a `\section*`).

Waiting Room: May Need for Reference Later or in Thesis

0.1. Most Important Stuff

- Okay, so first we are going to run a dataset for the full-horizon 16-week optimal plan with no rolling horizon (can we do this computationally?)
- Then, we'll compare it with how horizon length influences the optimality gap? I think will find a bit of a sweet spot between instability and computationally heavy that would be in that medium horizon range
- Then we can lock the times in for each planning horizon and see if it still produces reasonable results

- This will follow a matrix of combinations including planning horizons between two and sixteen weeks, re-optimization rolling-horizon windows between 1 day and 2 weeks, and disturbance levels ranging from no disturbance to severe.

0.2. What Needs to Happen Before This Can Be Helpful To People

- CARE/OCAP principles, all published results need to be generated with the open component and proper licensing that describes how it can be used
- Showing commercial solver against baseline solver in results to show still adequate but could benefit?
- Open software, not open data
- Licensing for code, docs and schemas
- RFI request process outlined clear
- (Jantti and Aho 2023) about IoT in forestry and connection to their discussion

(Lejeune and Kettunen 2017)

Problem Definition

- Operational-level, three-month planning horizon with daily/hourly/shift-level time-steps
- Decisions being made are machine scheduling and mobilization
- Set list of blocks, set list of equipment and workers and allocating/scheduling equipment and workers to blocks
- Objective functions (including possibly more flexibilty with that)

Each combination was replicated for science:

$$r = 1, \ldots, 30$$

to capture variability and distinguish between systematic and random variation. The replication using different random seeds allows us to account for the stochastic elements of disturbance in the model. This replication value is consistent with forest operations modelling (Beaudoin et al. 2008) and provides us with the necessary sample size to statistically analyze (enough data to do statistics with for sensitivity analysis, factor combinations, show error bars and confidence intervals, attribute findings to planning decisions, etc).

DESCRIPTION OF THE HARVEST PROCESS as it pertains to the operational environment. For any given site, each job can only be carried out by one type of machine. In other operations, some machines may be able to carry out more than one job (e.g. loader used for hoe-chucking and loading, processor used for processing and loading on the BC Coast); this would create a re-entrant problem in the model (Schulze et al., 2016 but not sure what this citation is located). For the purpose of this study, it can be assumed that each job is machine-specific. For each type of machine, the number of machines is limited. DESCRIPTION OF EQUIPMENT AVAILABLE TO CONTRACTOR. (table format helpful here). Expanding the scope of the problem from the block-level to the package-level (multiple blocks within the same working area), you also need to consider the mobilization of equipment into each block from the depot or from other blocks (or working areas) via low-bedding or walking (moving the machine without the assistance of a low-bed truck). Low-bedding is a faster process whereas walking generally moves the machine at a speed of 4 km/h (CITATION) and is only used when blocks are in close proximity to each other. (WALKING OPERATIONAL CONSTRAINT). There is also the difference between wheeled and tracked equipment to account for here. For example, a wheeled skidder could easily move from block to block whereas a tracked feller-buncher would likely require low-bedding if the distance exceeds economic feasibility.

To evaluate response to disturbance level and schedule robustness, we used the stochastic playback and disturbance models built into FHOPS, classified into three levels:

$$\ell \in \{\text{low}, \text{ medium}, \text{ high}\}$$

The same scenario without any disturbance introduced will serve as the baseline comparison with the three levels of disturbance intensity scenarios.

Validation and Benchmarking

- Calibration
- Validation
- Uncertainty, sensitivity analysis, robustness, agility

Validation in optimization models can follow the convention Janová et al. (2024) presented where there is face validation, additional validation (i.e. stress testing, robustness testing) and an explicit discussion as to how the model fulfills the stated purpose.

(Kazemisaboor et al. 2024) how rolling-horizon prevents against creaming

Harvest System Descriptions

Scheduling allocates resources to meet a pre-determined objective function. Scheduling can be used across a wide range of resources including time, money and machines. Machine scheduling is a commonly researched and practiced field, spanning many different industries, and has seen much evolvement alongside technological advancements (Martinelli et al. 2022). When it is completed in forestry, it falls within the operational planning level of hierarchical forest management. Machines make up a large part of forest operations, the day-to-day activities that allow us to practice sustainable forest management. The type of cutblock and its operational environment (terrain, piece size, etc) gives critical information as to which harvest system should be assigned to that cutblock. In British Columbia, different geographic regions rely on different harvesting systems as different harvesting systems are dependent on the piecesize, terrain and other attributes of the cutblock. The design and layout of the block would be specific to the harvest system and equipment used in the block and is therefore not a decision variable in this presented model. For models that support this type of decision-making, please see XXX. However, in terms of allocation of specific machines, there is still a need for scheduling to optimize especially when there are multiple machines of the same type available (i.e. like four feller-bunchers). With larger operations and a wider fleet of equipment, making the most out of this allocation becomes more necessary to ensure that all of this equipment is used efficiently and operations stay economically viable and feasible for the contractor and the licensee. In forest operations, the sequencing of different tasks is determined by the type of harvest system and category of machines. Each category of machines conducts dedicated tasks. For example, a buncher-skidder-processor combination is one allowed sequence. This is because they each complete different components of harvesting according to the proper precedence outlined in felling, primary transport, processing. Depending on the system, processing may occur before primary transport or after. In all cases and unsurprisingly, felling is required to be the first task in the harvesting stages. Models like Arora et al (2023) have tackled this using precedent relationships. This eliminates some complexity in the sequencing problem associated with forest operational machine scheduling, however, with different harvest systems allowing different sequences, there is still some complexity.

There is a difference between the coastal and interior operations in British Columbia, mainly a function of terrain and water access in these different areas. These can be grouped into three broad categories based on the primary transport used: ground-based, cable and helicopter. Ground-based operations tend to be more focused on gentle or moderate slopes although with the addition of winches (thus becoming winch assist operations or tethered operations), steeper slopes can be accessed. Steep slopes are also harvested using cable operations. Depending on the value of the timber, access and terrain, helicopters are also used. For the purpose of this model and its presentation as a generalized model, the most common harvest methods are depicted in Table 1.

The relationship of machines and productivity is mapped using productivity equations, that are based on block attributes. Depending on the category and type of machine, productivity may differ. Productivity equations are fed with a variety of factors including piece size, terrain characteristics (e.g. slope) among others. When designing the model, X and X were specified as factors to be included in the productivity equations given the prevalence and availability of this information across almost all blocks that would be harvested. Cruising summaries and basic terrain information is a fundamental input in block design and therefore, using this information in the productivity equations does not self-select or eliminate some blocks more than others from being included and assessed properly in the model. The model includes sequencing using absolute volume and area constraints.

(Frisk et al. 2016) example of a machine/worker scheduling model that needed to be decomposed into separate components (though not rolling horizon) so that it could be computationally tractable

- Testing what the model can and cannot do
- Scope and scale and depth of analyses for model
- Testing out the value of testing heterogenous system combinations that are sub problems which could actually be easier to solve than a super large number of the same system needing to be applied to a large number of blocks which would computationally be difficult to solve or a really large math number. This we'd be able to show the limits of optimality, time it takes and everything crunching the numbers. SO with the simpler systems being organized it would likely take a while longer than if we split it into sub problems to be solved, solving smaller problems within the general larger problem. This would be an interesting test to show the behaviour of the model rather than having 'good' practical implications
- Testing out pure systems separated vs mix of common harvesting systems (show that it works, does it change the solution or make it easier or more difficult) –> ideally, this would be easier to solve with sub-problems. More realistic and easier to solve is a win-win.

- Blocks

- Machines
- Landings

Block Data

- block id (unique block identification code)
- species mix
- area ha (area in hectares)

- Description of what a scenario must contain: block list with associated prescriptions, machines, shifts, roads, workers.
- Validation and unit tests to make sure the model isn't computing incorrect information
- Any needs for ethics approval
- Data sourcing and pre-processing requirements and steps (cleaning the data, sources, etc)

This paper presented a decision support tool for machine scheduling in forest operational planning.

- What was the strategy in designing this modelling framework?
- How were each of the solutions compared and validated against the old solutions?
- Evaluation method
- Validity check
- What datasets were used? What kind of data was fed into the model and how does this compare to realistic or relevant datasets?
- What are the improvements or limitations of this model compared to other options?
- What does this modelling framework offer that current solutions cannot?
- What is the trade-off between computing power, data availability (quantity vs quality)
- How does this feed into what the model does?
- What are the decision-makers/real-life operational decisions that will be impacted by the model findings or outputs?
- How will this knowledge be translated into real-world applications?
- What are the future research areas that could build off of this work?

0.3. Solver Performance

Okay, this could really be its own paper... hmmm do we add this to the full paper that Greg will write next year? (being early 2026)

As per the optimization literature, these four approaches will be evaluated in terms of runtime, convergence behaviour and scalability. Past studies have shown that ... about quality vs quantity when comparing solution methods.

- Mixed Integer Programming and Heuristic Solver Comparisons
- Comparison in runtime between MIP and heuristic approach for each of the different example case studies
- Time comparisons
- Computational capacity (runtime, memory, scale with different datasets)
- Model building time separate from model solving time

- Performance comparison between MIP and heuristic. Solving for both with measured gaps between referenced mathematical exact MIP and heuristic solution and then in discussion, discuss gap. When does breakdown happen (MIP), based on size of dataset?
- Push HiGHS as far as it can go to solve the MIPs (as the open-source solver). When does it stop working reliably (with reference to optimality gap?) over the same set of time (being control variable). Then, re-run same test with CPLEX or GuROBI and see if it can do better. Does HiGHS come undone faster or slower than GUROBI? What do you need to use in practice?
- Convergence mapping of MIP and heuristics for different sizes of datasets (runtime)
- Response to hyper parameter tuning built into the model. Do they have a 'good' or 'bad' performance? What are the SWOT of these? Can we solve production grade problems in a reasonable amount of time?
- Figure would show x axis of problem size (size of matrices) with baseline as the MIP solution and then plotted on the y-axis is the difference between the MIP and the heuristic (Z* score, percent gap) and you would comment on shape of the curve (linear, exponential, slow or fast gap growing) and showing where MIP dies

The model includes four solution methods. The first is a mixed-integer linear programming model (MIP) that utilizes HiGHS as a default solver. If you run into the paywall, you can also use Gurobi but it is expected that the free open-source solver would be usable for the scale of problems. MIPs are commonly used for solving machine scheduling and allocation problems because it can show discrete decisions and constraints in mathy language. MIP models solve to optimality. HiGHS is the default solver and has been able to show comparable results to industry-standard solvers. However, when the problem size exceeds the computational capabilities of HiGHS, Gurobi is also an option (citation).

In addition to the optimization method, there are also three metaheuristics included as solution methods. Simulated annealing (SA), tabu search (TS) and iterated local search (ILS) are all hill-climbing based approaches that can explore large solution spaces. They are generally used when computational availability or time is limited or when the MIP becomes too large to be solved to optimality.

(Boston and Bettinger 1999) for the comparison between tabu search, simulated annealing and an integer programming model

- Simulated annealing explanation (Kirkpatrick et al. 1983) and discussion of benefits/use (Nikolaev and Jacobson 2010). Simulated annealing uses a temperature schedule to escape local minima, mimicking the annealing method used in metal production.
- Tabu search explanation (Glover et al. 1993). It puts short-term bad solutions on the no-fly list so it doesn't keep going back to them so it can search the solution space faster.
- Iterated local search explanation (Lourenço et al. 2003) avoids local optima. It is best suited for problem type X.

The three used heuristics are simulated annealing (SA), tabu search (TS) and iterated local search (ILS).

- What are the strengths and limitations of these three heuristics?
- What is the ability or limitations of these in using the models? What gets us a

wider available range of solutions?
- How do these compare with the MIP? Here, introduce Beaudoin et al. (2008) and discuss anticipation and the limits of optimality
- Where do the MIPs fail (threshold of the MIP)

0.4. Sensitivity and Uncertainty

- Sensitivity analyses for different input parameters
- Model drivers
- Radar plots
- Input sensitivity
- Violin plots for uncertainty mapping (using KPIs)

Developing this model lays the groundwork for complementary research with open-source trucking logistic models, permitting future research that can investigate the impact of harvesting decisions on transportation and vice versa. For example, how does the spatial distribution of harvesting impact transportation costing or route scheduling? Inversely, additional studies could explore the impact of different trucking and fleet configurations (e.g. percent electric, payload limitations) on optimal harvest scheduling design. This collective knowledge could lead to industry-wide advancements across different forest operations with greater agility and lower barrier-to-entry. This project is an initial step towards framework development that can function within the complex dynamic system of forest operational planning.

The open-source modelling framework is developed according to best practices and methodology outlined in Lamprecht et al. (2020) to create FAIR (findable, accessible, interoperable and reusable) software. Some of these components include:

- Documentation
- Publicly available primary data (collect input data into sets and lists of inputs for model)
- Metadata available and description on README explaining in plain-language how to use the tool
- Pre-processing and processing of data made clear and coded, published for reproducibility transformed into secondary data (model input)
- Automation with more public, openly public scripts of raw output to automate analysis and number-crunching of output
- Hosting of the framework and resources in a public repository (GitHub) with development plan for maintenance and version tracking
- Interoperability
- Using a recognized open-source license (MIT), clearly stated
- Citation file to make sure it can be cited

0.4.1. Mathematical Formulation

0.4.2. Scope

0.4.3. Sets and indices

0.4.4. Decision variables

Insert a figure here that denotes the different decision variables and their relationships with each other. Pull from other formatting in papers that you found easy to read.

0.4.5. Constraints

Detailed description of the different constraints, pointing to each of the associated equations and making them make sense in a real-life use.

0.4.6. Objectives

The objective function of this model is to minimize/maximize some goal. Short description of what this means and relevance.

The objective of the model is to generate an optimal operational-level schedule (three-month planning horizon with shift-level time steps) for a set of cut blocks using available equipment and labour. The mathematical formulation of the proposed model is as follows:

$$\text{min or max} \quad \sum_{t \in T} \sum_{m \in M} \sum_{b \in B} \sum_{p_r \in P_R} \sum_{p \in P} c_{tmbr_p} x_{tmbr_p} \tag{1}$$

s.t.

0.5. Model Architecture

The model architecture is developed with a modular structure. There is a CLI and an API.

- Core
- CLI
- Data
- Evaluation
- Model
- Solve

The planning horizon of operations can vary and we would like to use this model to explore what the bounds are of how short and far ahead you need to plan for forest operational planning in both math/computational capacity and also just The questions of this paper are:

- How far do you need to plan before the forecasted plan goes bad?
- What rolling-horizon length produces 'stable' schedules?
- Do different horizon lengths converge on the same results?
- Does reoptimizing every week outperform longer plans with less reoptimization?
- Do you need to go to heuristics to solve these problems?
- Does mixing harvesting systems affect problem solving?

0.6. Introduction Thoughts

Although the scheduling problem has been repeatedly solved in site-specific contexts, the lack of open and reusable tools has resulted in fragmented progress within the field. There is no need to reinvent the wheel but there is a need to make sure the wheel can be reproduced robustly and that the wheel remains accessible. Why has operational planning been fragmented? Part of this is from the computational complexity of the underlying planning problem.

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